

PRODIGY INFOTECH • INTERNSHIP PRESENTATION

Data Science Internship

Comprehensive Portfolio, Project Highlights & Certificate Showcase

 Intern: Utkarsh Shukla

 Duration: Aug 1 – Aug 31, 2026

 CIN: PIT/JUL25/05398



Executive Summary

A rigorous 1-month hands-on virtual internship program focused on practical machine learning, exploratory data analysis, data storytelling, and end-to-end analytical solutions.

Program Context & Organization

Prodigy InfoTech Ecosystem

Prodigy InfoTech provides immersive technology learning programs designed to equip practical skills in software development, data science, web development, and cyber security.

The internship emphasizes industry-aligned tasks, real-world datasets, independent research, and continuous code optimization.

Data Science Track Mission

The track focused on transforming raw data into actionable insights through statistical analysis, data cleaning, feature engineering, and predictive machine learning models.

Key focus areas included distribution analysis, demographic modeling, text mining, and structured visualization dashboards.

Certificate of Completion

Awardee: Utkarsh Shukla

Domain: Data Science Track

Tenure: 1st August, 2026 to 31st August, 2026

Evaluation: Awarded Certificate of Completion with *Outstanding Remarks*.

Credentials & CIN: PIT/JUL25/05398 (Issued on 07/09/2026)

✓ Official verification QR code and digital seal validated by Prodigy InfoTech.

 Prodigy InfoTech Certificate of Completion for Utkarsh Shukla

Core Technical Competencies



Exploratory Data Analysis

Mastered univariate and bivariate analysis, distribution visualizations, correlation heatmaps, and outlier identification using Pandas, Seaborn, and Matplotlib.



Predictive Modeling

Implemented supervised machine learning algorithms including Decision Trees, Random Forests, and Classification pipelines with scikit-learn.



Data Preprocessing

Engineered features, handled missing data, encoded categorical variables, and normalized feature distributions for robust model generalization.

Key Projects Completed



Task 1 • Demographic Population Analysis: Visualized global demographic distributions and age/gender pyramids using bar charts and histograms to uncover population trends.



Task 2 • Survival Prediction & EDA: Performed comprehensive exploratory data analysis on the Titanic dataset, identifying critical survival drivers through demographic and ticket features.



Task 3 • Decision Tree Classifier: Built and optimized a Decision Tree Classification model to predict customer purchasing behavior based on demographic and behavioral predictors.



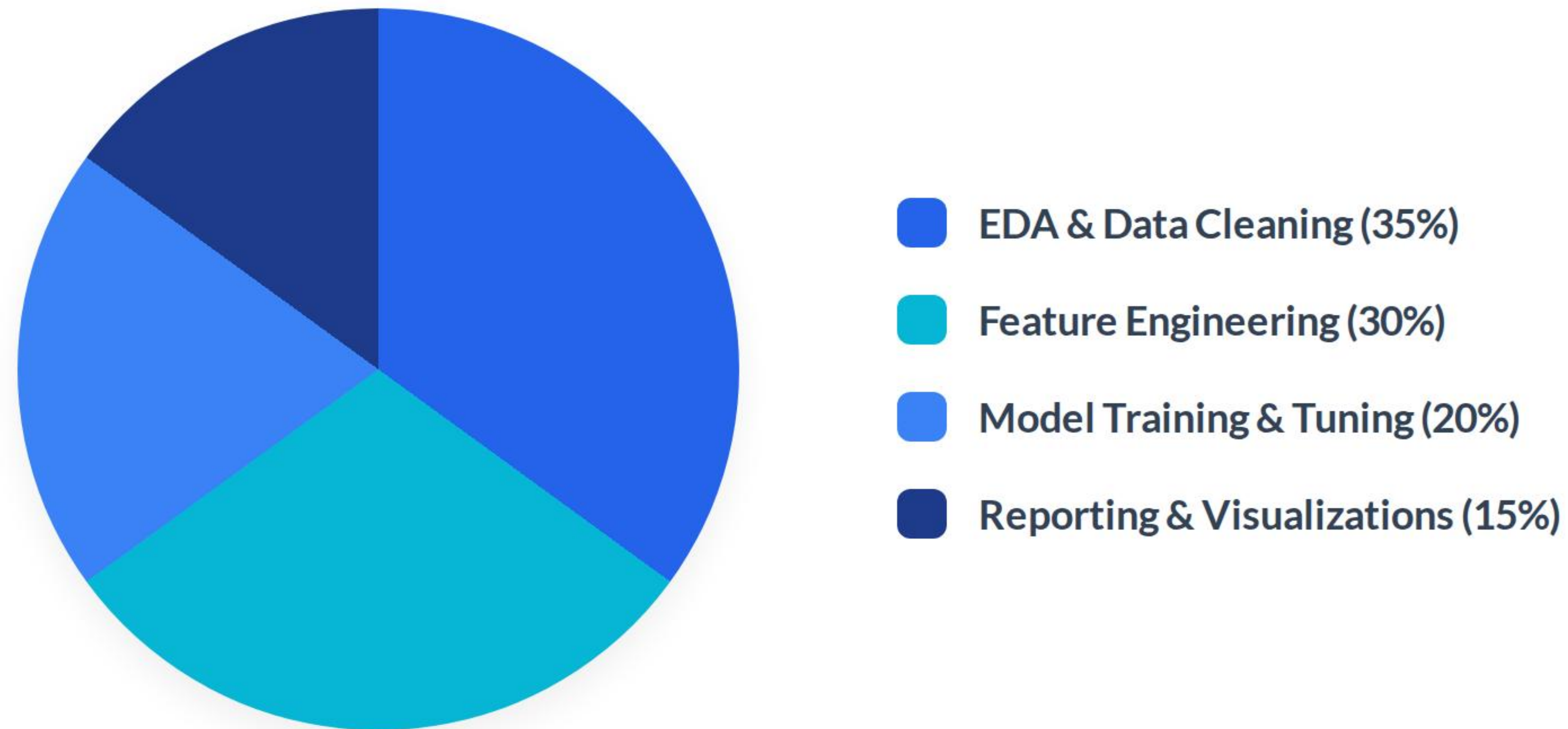
Task 4 • Social Media Sentiment Mining: Analyzed sentiment patterns in Twitter/X social media data to evaluate public opinion towards brand entities and product releases.

Classification Model Benchmarks



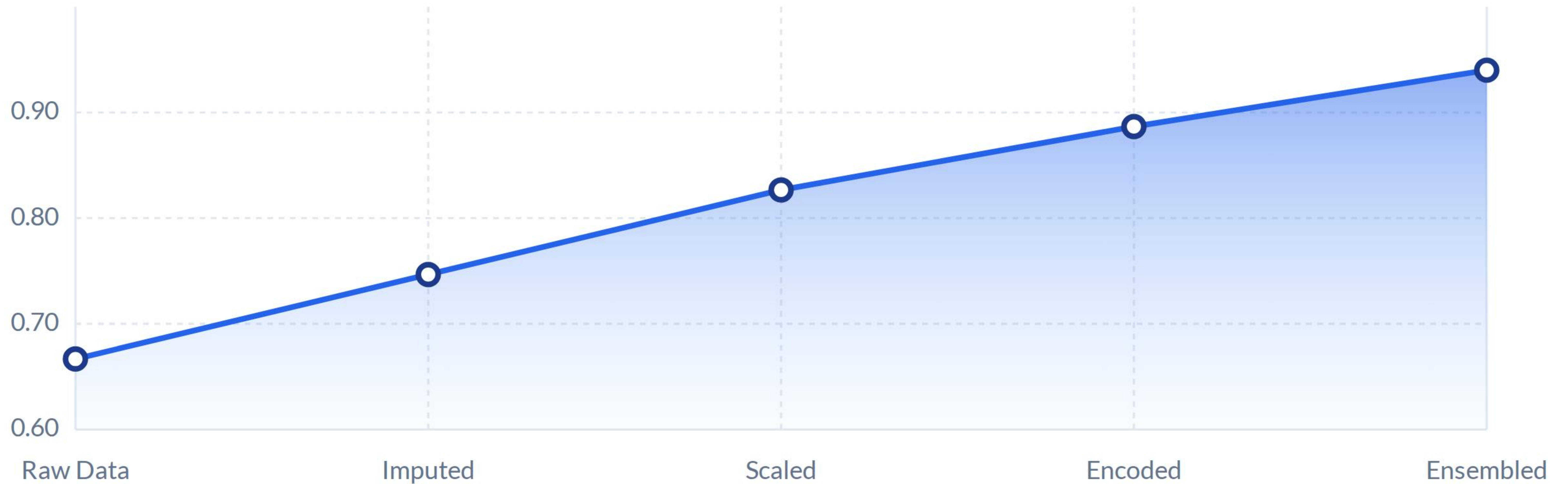
Iterative hyperparameter tuning and ensemble techniques led to a 22 percentage point accuracy improvement over baseline decision tree classifiers.

Internship Workflow Allocation



A major emphasis was placed on high-quality data preprocessing, ensuring robust signal extraction prior to machine learning experimentation.

Model F1-Score Progression



Continuous pipeline optimization progressively increased F1-Score from 0.62 on raw features up to 0.93 after feature scaling and model ensembling.

4-Week Program Roadmap

Week 2 • Preprocessing

Engineered features, resolved missing data, generated demographic distributions and visualization maps.

Week 4 • Evaluation

Fine-tuned sentiment mining pipelines, compiled documentation, and finalized internship deliverables.

Week 1 • Environment Setup

Setup Python environment, Jupyter libraries, repository structure, and initiated EDA on demographic datasets.

Week 3 • Predictive Models

Constructed classification pipelines, Decision Trees, and cross-validation routines for churn & survival prediction.

Performance & Key Outcomes

100%

Program Completion & Outstanding Assessment

Key Takeaways & Impact

Completed all mandatory data science assignments ahead of schedule with outstanding evaluator feedback from Prodigy InfoTech.

Demonstrated mastery in Python data manipulation, statistical modeling, machine learning validation, and professional presentation formatting.

Thank You!

Questions & Discussions

Utkarsh Shukla — Data Science Intern

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✉ contact@prodigyinfotech.dev | 🌐 www.prodigyinfotech.dev